

Kühni end sections

Preliminary mechanical proposal

PRELIMINARY — NOT FOR FABRICATION. A dimensioned proposal is selected below for engineering development. Capture performance, actual required end heights, pressure design and release remain HOLD. Selected allowances are not measured requirements or published universal clearances.

Proposed configuration: top 700 mm ID × 1200 mm straight → unchanged active 700 mm ID × 4200 mm → bottom 700 mm ID × 300 mm feed neck + 173.205 mm conical transition + 900 mm ID × 1200 mm straight.

Bottom 900 mm is selected for an **engineer-selected 200 µm RRBO capture screen**, with **$U \leq 0.50 v_t$** and **no coalescence growth credit**. Calculated minimum ID is **868.75 mm** at the worst bottom flow proxy. This is not proof that physical enlargement is necessary, that the real population has this cutoff, or that carryover is acceptable.

1 · What is preserved, and what is selected

Classification	Basis / decision
INHERITED, unchanged	Active ID 700 mm; active H 4200 mm; 20 compartments × 210 mm pitch; rotor 231 mm; stator free-area ratio 0.4; 30 rpm; adopted power number $N_p=1.2$. Physical $N_T=7$ and efficiency 0.35 are not changed.
ENGINEER-SELECTED	Bottom capture target 200 µm; velocity fraction 0.50; 900-mm bottom; 30° cone half-angle measured from axis; all end-zone heights, interface levels, nozzle centre elevations and 0.5-m/s nozzle screening velocity below. None is a vendor guarantee.
CALCULATED	Finite-Re terminal speeds, inverse capture cutoffs, superficial velocities, 868.753-mm required ID, 173.205-mm cone rise, nozzle bulk velocities and dimension chains.
NOT ESTABLISHED	Actual incoming end-zone drop populations, dispersed load, acceptable carryover, equilibrated outlet properties, coalescence/band capacity, nonuniform-flow allowances, surge duty and mechanical suitability.

The previous report withheld geometry selection because the actual bottom population was unknown. This proposal advances a *disclosed provisional design duty*; it does not overturn that evidence limitation. Neither 250–253 µm nor 200 µm is literature-proven conservative for RRBO/NMP. A 150-µm duty would require about 1142.58 mm by the same screen (1200-mm candidate), and 100 µm about 1696.60 mm. A consequential finer tail may defeat any selected duty.

Unbound outlet-flow envelope, not Stage-2 approval

Unbound scenario	Raffinate kg/h	Top Q proxy m ³ /h	Extract kg/h	Bottom Q proxy m ³ /h
4	3321.139186	3.821794230	2240.460814	2.207350556
7	3278.673563	3.772927000	2282.926437	2.249188608

N4 controls the top flow; N7 controls the bottom flow. Taking separate maxima is an explicit design envelope, not a single thermodynamic operating state. Stage-2 job 145e962e-892a-4424-aab9-91532af9f5ef remains unbound; releaseEligible=false and Stage-4's Stage-2 job link is NULL. The source hash differs from current Stage 1 despite matching audited input fields. Physical $N_T=7$ does not silently bind N7 outlet results.

RRBO feed: $4000 \text{ L/h} \times 869 \text{ kg/m}^3 = 3476 \text{ kg/h}$. Wet solvent: $S/O=0.6 \text{ mass} = 2085.6 \text{ kg/h} = 2054.778325 \text{ L/h}$, 99.5/0.5 wt% NMP/water. Total mass closes at 5561.6 kg/h in both scenarios. Product Q = product mass / nominal phase density; these are **volume proxies**, not measured mixture volumes. Equilibrium dissolved NMP in raffinate and hydrocarbon in extract are not physical droplet entrainment loads.

2 · Physics and reproducible diameter comparison

Nominal 40 °C property	Bottom	Top
Continuous phase ρ / μ	NMP-rich: 1015 kg/m ³ / 0.001416 Pa·s	RRBO-rich: 869 kg/m ³ / 0.0598 Pa·s
Dispersed phase ρ / μ	RRBO: 869 kg/m ³ / 0.0598 Pa·s	NMP: 1015 kg/m ³ / 0.001416 Pa·s
Common $ \Delta\rho / \sigma / g$	146 kg/m ³ / 0.011 N/m / 9.80665 m/s ²	Same nominal proxies
Dispersed / continuous viscosity ratio	42.2316	0.0236789

$Re = \rho_c v_t d / \mu_c$; $CD = (24/Re)(1 + 0.15 Re^{0.687})$
 $CD \rho_c v_t^2 / 2 = (2/3) d |\Delta\rho| g$; $U = Q/A$; $A = \pi D^2/4$
 Criterion: $U \leq f v_t$, $f=0.50$; $D_{req} = \sqrt{4Q/(\pi f v_t)}$
 Inverse cutoff: $v_t(d_{crit}) = U/f = 2U$; net return speed = $v_t - U$.
 $E_o = |\Delta\rho| g d^2 / \sigma$. All roots use bracketed bisection.

Bottom: high viscosity ratio suppresses internal mobility; finite-Re spherical Schiller-Naumann is an appropriate *isolated-drop screening approximation* at the small cutoffs. Over the requested diameter range $Re \approx 0.110$ – 0.588 and $E_o \approx 0.0026$ – 0.0083 : static deformation is negligible as a leading correction on this basis. This does not exclude turbulence, swarm retardation or property errors. Using RRBO viscosity as the continuous drag viscosity at the bottom would be wrong.

Stokes and clean Hadamard-Rybczynski are diagnostics, not the adopted finite-Re calculation. At 250 μm , Stokes overpredicts SN speed by 10.21%; HR/Stokes differs by only 0.777%. Myint et al. Eq. 9 gives a clean-to-SN correction of about 0.730% there, but E_o is below its tested minimum 0.017. This is an extrapolation diagnostic, not a new validated RRBO/NMP model. No bubble deformation correlation is substituted [R1].

Bottom: RRBO rising against downward NMP-rich flow

Paired entries are **N4 / N7**; $Q = 2.207350556 / 2.249188608 \text{ m}^3/\text{h}$. d_{crit} is a required size at equality to the assumed margin, not a predicted population or the zero-net-rise diameter.

ID mm	U mm/s N4 / N7	$d_{crit} \mu\text{m}$ N4 / N7	200- μm screen N4 / N7
700	1.593246 / 1.623444	250.031 / 252.626	FAIL / FAIL
800	1.219829 / 1.242950	216.265 / 218.461	FAIL / FAIL
900	0.963816 / 0.982084	190.720 / 192.628	PASS / PASS
1000	0.780691 / 0.795488	170.682 / 172.371	PASS / PASS
1200	0.542146 / 0.552422	141.198 / 142.574	PASS / PASS

Top: NMP settling against upward RRBO-rich flow

Paired entries are **N4 / N7**; $Q = 3.821794230 / 3.772927000 \text{ m}^3/\text{h}$. All top cutoffs are **conditional SN diagnostics, not validated liquid-drop capture predictions**.

ID mm	U mm/s N4 / N7	Conditional dcrit mm N4 / N7
700	2.758537 / 2.723265	2.080748 / 2.066822
800	2.112005 / 2.085000	1.811332 / 1.799330
900	1.668745 / 1.647407	1.604444 / 1.593885
1000	1.351683 / 1.334400	1.440405 / 1.430972
1200	0.938669 / 0.926667	1.196472 / 1.188687

Retain the 700-mm top as a candidate, not an approved settler. The saved active-compartment mean is 6.068312 mm **NMP-in-RRBO**, never RRBO-in-NMP. SN gives $v_t=36.665670 \text{ mm/s}$, $Re=3.2333$, $E_o=4.7931$ and $We=0.6445$. Significant deformation/mobility uncertainty remains. The 700-mm critical size is about 2.08 mm ($E_o \approx 0.56$); mean d_{32} exceeding it does not prove fine-tail capture. Top viscosity ratio is below Myint's tested lower limit 0.1. No coalesced mean or coalescence-growth benefit is assumed.

3 · Selected bottom duty and spatial height basis

200-μm design screen, worst bottom Q	Calculated result
Terminal speed / Re / Eo	2.108000 mm/s / 0.302206 / 0.005206
Allowed U = 0.50 vt	1.054000 mm/s
Required area / ID	0.592765 m ² / 868.753 mm
Selected ID / area / actual bulk U	900 mm / 0.636173 m ² / 0.982084 mm/s
Margin ratio / excess above assumed criterion	1.073228 / 7.32%
Net upward speed at selected ID	1.125916 mm/s

900 mm is the smallest of the evaluated 700/800/900/1000/1200-mm sizes passing this assigned 200-μm duty. Its modest excess capacity is not a demonstrated allowance for jets, fouling, swarm effects or property uncertainty. At 700 mm, 200-μm net rise is positive but the 50% criterion fails; at 150 μm the net direction is downward. More height cannot repair either a failed area criterion or a wrong-direction trajectory.

Heights are selected layout allowances, not an arbitrary residence-time result

$t_{\text{return}} = L/(v_t - U)$, if $v_t > U$; $t_{\text{bulk}} = AH/Q = H/U$.
If $L=H$, $t_{\text{return}}/t_{\text{bulk}} = U/(v_t - U)$, independent of height.

Therefore these equations cannot determine a unique sufficient end height. The proposal instead reserves identifiable physical spaces and sets elevations. **The numeric allowances below are engineering selections for layout development, not source-prescribed or calculated necessary clearances.** Actual hydraulic height remains HOLD until inlet decay, band/coalescence, withdrawal and controls are qualified [R2-R6].

Region (bottom upward)	Selected axial allowance	Purpose / qualification
900-mm withdrawal zone	300 mm	Outlet centre 200 mm above lower tangent; anti-vortex device and accessible drain below. Local drawdown check required.
900-mm clear return zone	600 mm	Space below the lowest interface-control elevation, above withdrawal zone. No claim that 600 mm dissipates a particular jet.
900-mm interface/control reserve	200 mm	Normal interface ± 100 mm. This is a reserved combined operating envelope, not a predicted 200-mm dispersion band. If band extent plus excursions exceeds it, increase/reposition layout.
900-mm upper guard	100 mm	From highest reserved interface elevation to cone lower end; verify interface cannot invade transition.

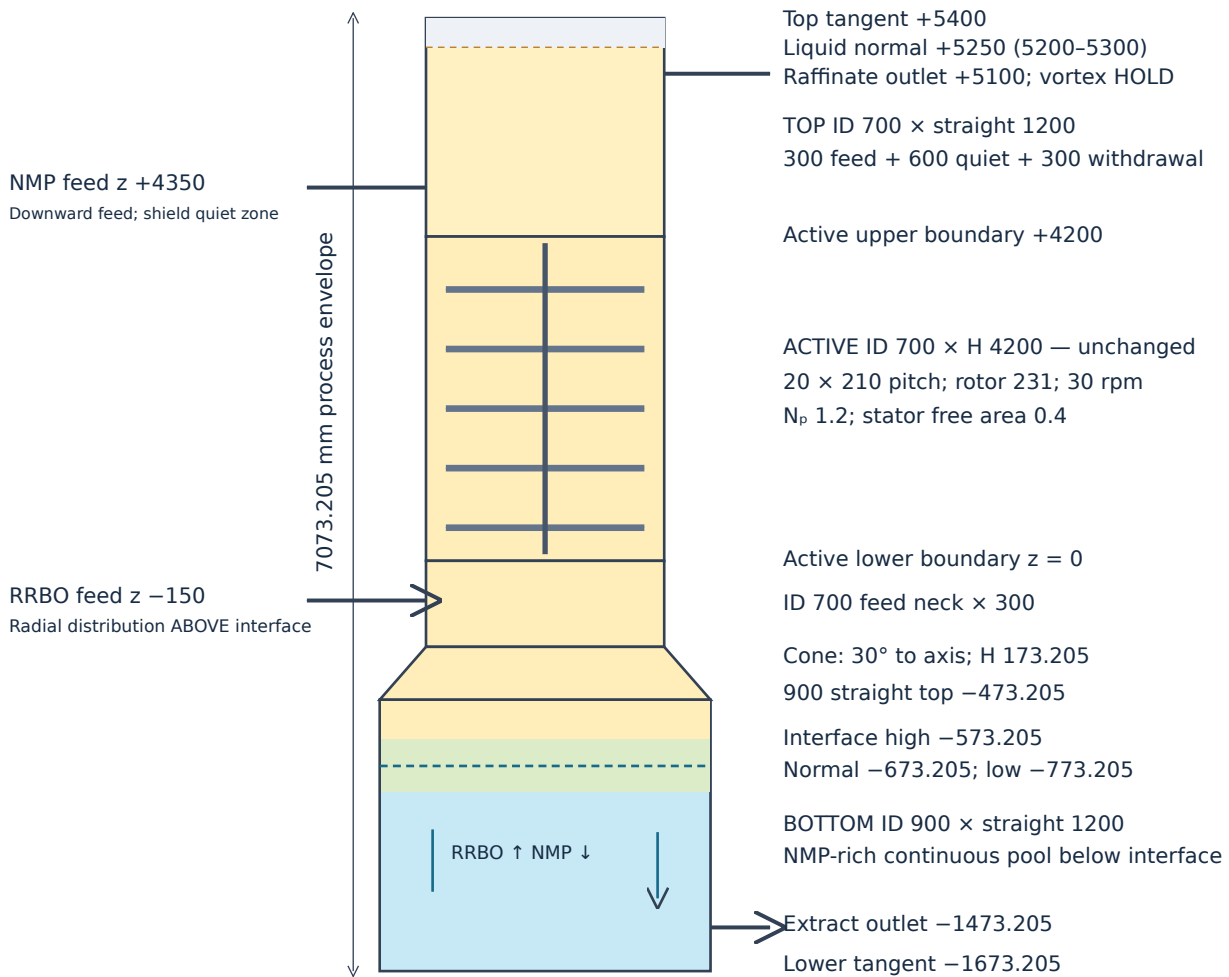
Region (bottom upward)	Selected axial allowance	Purpose / qualification
900-mm straight total	1200 mm	300 + 600 + 200 + 100. Mechanical layout selection only.
900→700 cone	173.205 mm	Radial change $(900-700)/2 = 100$ mm; $H = 100/\tan 30^\circ$. Angle chosen, height geometrically derived.
700-mm feed neck	300 mm	Separate RRBO feed/distributor from active boundary and bottom interface; no existing active internals moved.

At the normal interface, outlet-centre-to-interface distance is 800 mm (range 700–900 mm). For the hypothetical uniform 200- μ m trajectory, $t_{\text{return}}=710.5$ s and $H/U=814.6$ s for the *same* 0.8-m slab. Across the 600-mm quiet allowance the corresponding times are 532.9 and 610.9 s. These are illustrative transit comparisons only: drops near a radial outlet may escape much sooner, and no actual release elevation is established.

Normal cylindrical NMP pool inventory below interface, excluding head, is 0.636173 m³; full 200-mm control-reserve volume is 0.127235 m³. Do not credit all of this as usable surge: dispersion, internals, alarm margins and level uncertainty reduce it. Required surge volume must come from $\Delta V = \int (Q_{\text{in}} - Q_{\text{out}}) dt$ for an agreed upset and response time; neither is supplied.

4 · Dimensioned phase-consistent arrangement

LAYOUT REFERENCE • all elevations mm • not to scale



Amber: RRBO continuous. Blue: NMP continuous. Green: interface-control reserve, NOT a predicted dispersion band.

Reference datum $z=0$ is the preserved active lower boundary; $z=4200$ is its upper boundary. The drawing is schematic: no head shape, weld detail, support, plate thickness fit-up or pressure-shell design is implied. Terminal stator plane/face offsets (including 4-mm plate thickness) must be reconciled to current CAD; no stator is relocated by this report.

Bottom flow topology and interface protection

The dense NMP-rich continuous pool is **below** the bottom liquid-liquid interface. RRBO is continuous above that interface through the cone, feed neck and active section. Descending NMP drops merge into the lower pool; entrained RRBO in that pool rises to the interface and rejoins the upper RRBO phase. This final interface-merger process still needs coalescence capacity, even though no drop-growth credit is used in the screen.

RRBO feed at $z = -150$ lies in the 700-mm neck, **above the highest interface** by 423.205 mm. Use a low-momentum circumferential/radial distributor and shield so feed is directed toward the active entrance, not as an axial jet onto the interface. No bulk RRBO feed is injected into the NMP pool. There is no forced extract-return pipe or assumed pump: gravity phase exchange uses the open process cross-section. Distributor holes, open return area, shaft obstruction and pressure drops need design; none is assigned fictitious capacity.

The neck is 300 mm deep, but feed centre is only 150 mm from the active lower boundary. This is **not** a claim of 300-mm distributor-to-stator clearance. The lowest rotor/stator agitation-decay region must be checked independently. The highest interface is 573.205 mm below the active boundary; whether this is sufficient is unvalidated.

5 · Top layout, nozzles and control philosophy

Top region	Elevation range mm	Selected allowance / function
Feed / terminal-stator buffer	4200 to 4500	300 mm; NMP distributor at 4350, directed downward toward active entry.
Quiet return path	4500 to 5100	600 mm; descending NMP drops have an unobstructed path back to active region.
Withdrawal / liquid-level space	5100 to 5400	300 mm; raffinate outlet centre 5100; liquid normal 5250, reference low/high 5200/5300.
Top total	4200 to 5400	1200 mm, engineer-selected straight section; no diameter transition.

Use a distributor/turbulence shield that separates fresh NMP injection from the upper quiet zone while permitting upward RRBO and downward separated NMP to pass. The distributor centre is 150 mm above the active upper boundary; the buffer is 300 mm in total. Neither is an approved last-stator/rotor calming clearance. Supports must not impinge on frozen internals. A blind plate that blocks countercurrent return is unacceptable.

The top has an RRBO-rich liquid/gas level, **not an invented second stable liquid-liquid interface**. At the reference normal level, the raffinate outlet centre is submerged 150 mm (100 mm at low level). For a 70-mm bore, the opening crown has only 65-mm cover at low level; anti-vortex/entrainment checks may require a lower outlet or higher level. Minimum straight-section freeboard is 100 mm at high level; this is not a qualified gas-disengagement/relief allowance. Head space is additional and unspecified.

Over the selected 600-mm top quiet path, the conditional mean-drop return time is 17.7 s versus 217.5 s bulk transit. At the conditional top cutoff, these are equal under the 0.5 screen. This illustrates the algebra; it does not establish top capture or justify 600 mm as a hydraulic minimum.

Retain existing 70-mm process bores provisionally

The previous proportional 70-mm bores can be retained as preliminary candidates rather than inventing a standard “DN70.” The selected bulk-nozzle velocity screen is **0.5 m/s** (engineering choice, not a breakup or pressure-loss guarantee). Minimum bore = $\sqrt{[4Q/(\pi \times 0.5)]}$. Nominal DN and schedule must subsequently provide the actual bore.

Service	Q m³/h	Centre z mm	70-mm bore velocity m/s	Min bore at 0.5 m/s, mm
RRBO feed	4.000000	-150.000	0.288716	53.19
Wet NMP feed	2.054778	4350.000	0.148312	38.12

Service	Q m ³ /h	Centre z mm	70-mm bore velocity m/s	Min bore at 0.5 m/s, mm
Raffinate outlet (N4 envelope)	3.821794	5100.000	0.275854	51.99
Extract outlet (N7 envelope)	2.249189	-1473.205	0.162344	39.89

All four bulk velocities are below the selected screen. Pipe friction (especially viscous RRBO), allowable nozzle loads, control-valve pressure loss, startup flow and distributor-hole jet velocities are unqualified. A low inlet-pipe velocity does not prove low local shear. Bottom outlet opening crown is 665 mm below the lowest reference interface; provide an anti-vortex arrangement without blocking droplet return and verify drawdown.

- **Bottom:** continuous interface measurement plus independent high/low interface alarms; conceptual interface LIC modulates extract withdrawal. Measurement technology must work with this density/dielectric contrast and dispersion/crud. Alarm/trip positions, response time and fail actions require HAZOP; reference operating levels are not certified trip settings.
- **Top:** liquid level measurement/alarms and raffinate withdrawal control keep the active section flooded. Pressure control/blanket, vent and independent relief require confirmed pressure basis, gas/vapor duty and code review. Saved operating pressure field is "2.0"; confirm gauge/absolute interpretation before mechanical use. Do not equate operating pressure to design pressure.
- **Auxiliaries:** reserve low-point drain/flush, top vent/relief, representative samples at both product nozzles and interface instrumentation connections. Existing 28-mm vents/drains and 17.5-mm generic sample/instrument ports are references only, not qualified sizing. Specify their duty, compatibility, plugging risk and maintenance access before choosing final bore/DN.

6 · Elevation schedule and release boundary

Feature	z mm / dimension	Authority
Active lower / upper boundary	0 / +4200	Inherited active envelope; terminal-plate offsets HOLD
Top straight upper end / reference tangent	+5400	Selected layout; head beyond this
RRBO feed / wet-NMP feed	−150 / +4350	Selected nozzle centres
Raffinate outlet / top normal liquid	+5100 / +5250	Selected, submergence HOLD
Top low / high liquid	+5200 / +5300	Reference control envelope, not trips
700-mm neck lower end	−300	Selected 300-mm neck
Cone lower end / 900-mm straight upper end	-473.205	Calculated from selected angle/IDs
Bottom high / normal / low interface	-573.205 / -673.205 / -773.205	Selected reserve, not measured band
Extract outlet centre	-1473.205	Selected 200 mm above lower tangent
900-mm straight lower end / tangent	-1673.205	Calculated dimension chain
Extra bottom length	1673.205	300 + 173.205 + 1200
Process straight/transition envelope	7073.205	4200 + 1200 + 300 + 173.205 + 1200

7073.205 mm is not the overall fabricated vessel height. It excludes heads, nozzle projections, flanges where they add length, drive, shaft seal, support feet/skirt and maintenance lifting space. Heads and supports require a fresh mechanical calculation for the enlarged bottom. Cone angle does not establish stress acceptability. Pressure/vacuum design, material/corrosion allowance, shell/head thickness, reinforcement, vibration, shaft support and access remain open. The existing Stage-5 5950-mm vessel / 7000-mm overall reference is not updated.

Item	Present disposition	Evidence needed to release
Active section	PRESERVED, not re-optimized	Existing active validation obligations remain; no new operating approval.
Bottom 900 × 1200 + neck/cone	PRELIMINARY SELECTED LAYOUT	Actual RRBO-in-NMP volume-weighted DSD and entrainment load; acceptable dispersed carryover; outlet-mixture properties and continuous/pilot separation evidence.
Top 700 × 1200	PRELIMINARY RETAINED CANDIDATE	NMP-in-RRBO lower-tail distribution; applicable deformation/mobility model or settling measurements and carryover target.

Item	Present disposition	Evidence needed to release
End height/interface bands	LAYOUT REFERENCE ONLY	Batch settling/coalescence kinetics, band height versus load, inlet/rotor disturbance decay, outlet trajectories, surge/control envelope.
Outlet flow envelope	UNBOUND SCENARIO SCREEN	Formally compatible downstream boundary binding; do not request re-entry of already available trial data.
Nozzles / distributor / return area	PROVISIONAL 70-mm bulk bores	Pressure loss, local jets, open areas, vortex protection, piping/instrument/vent duties.
Pressure vessel / release	HOLD — NOT FOR FABRICATION	Code/mechanical design, controls/HAZOP, vendor hydraulic approval and controlled drawing revision.

Next evidence should test representative equilibrated RRBO/NMP at 40 °C, including water, trace contaminants, aging/solids and worst operating condition. Sampling must not manufacture a larger drop population. Representative batch video/settling curves and continuous/pilot or vendor confirmation should establish sedimentation and interface-coalescence capacity separately. If interface merger cannot match incoming dispersed volume, band inventory grows despite a dilute velocity-screen pass.

7 · Verified source basis and limitations

This synthesis uses the retrieved evidence documented in `disengager-investigation-literature.md` and `disengager-drag-evidence.md`. No additional source is claimed to have been fetched for this proposal. The cited literature supplies mechanisms and model context, **not the chosen 200- μm duty, 50% factor, 300/600/200/100-mm allowances or 30° cone angle.**

1. **[R1] Myint, Hosokawa & Tomiyama (2006), “Terminal Velocity of Single Drops in Stagnant Liquids,” JFST 1(2), 72-81.** [Publisher PDF](#); DOI 10.1299/jfst.1.72. Full paper retained. Definitions Eqs.1-3 p.72; HR Eq.4 p.73; liquid-drop Eq.9 and SN Eq.10 p.77; applicability pp.79-80. Experimental bounds: $-11.6 < \log_{10} M < -0.9$, $0.17 < Re < 200$, $0.017 < Eo < 12.1$, $0.1 < \kappa < 100$. Reported error is not an RRBO/NMP uncertainty guarantee.
2. **[R2] Sulzer, Liquid-liquid separation technology.** [Full manufacturer brochure](#). Introduction pp.2-3/Fig.1: separate drop-drop and drop-interface coalescence; “Feed Inlets and Calming Baffles,” p.9: equalization and disturbance control; coalescers pp.10-12. Supports mechanisms and optional aided separation, not universal numerical clearances or bare-gravity 250- μm acceptance.
3. **[R3] Henschke, Schlieper & Pfennig (2002), CEJ 85, 369-378.** [Publisher abstract/introduction/section previews](#); DOI 10.1016/S1385-8947(01)00251-0. Batch determination of a coalescence parameter; film drainage and interfacial coalescence. Full paper not obtained; no degraded-preview equation imported.
4. **[R4] Hartland & Jeelani (1987), CES 42(8), 1927-1938.** [Publisher abstract](#); DOI 10.1016/0009-2509(87)80139-2. Sedimentation/band/coalescence dependence and batch-to-continuous model selection; full text not obtained. No fixed band height transferred.
5. **[R5] Schäfer, Hlawitschka & Bart (2022), CJCE 100(9), 2331-2346.** [Open article](#); DOI 10.1002/cjce.24503. §3, Fig.8 and §3.1: gravity-settler droplet behavior, vertical countercurrent head settlers and disturbance control. Mesh/horizontal laboratory dimensions are not this vessel’s sizing rules.
6. **[R6] Pfennig group, University of Liège.** [Coalescence, liquid-liquid phase separation and settlers](#), introductory technical discussion: trace components and representative lab tests. [Leleu & Pfennig, ISEC 2022, slides 31-33](#): polydispersity/lag/swarm/settler concept; not RRBO/NMP data.
7. **[R7] Sulzer OptimEXT extraction brochure.** [Manufacturer PDF](#), overview/extractors, ECR and ECP distributor sections. Phase orientation and distribution matter. [ECR product page](#). Neither retrieved source supplies a numerical ECR end-height/capture rule; no specific optional ECR-end coalescer arrangement is attributed to them.
8. **Comparator only:** [SkimOIL API-style oil/water separator brochure](#), “Per API guidelines design criteria / Oil droplet size.” Its 150- μm refinery-wastewater convention is not direct verification of normative API 421 text and is not transferred to RRBO/NMP.

No verified source proves 250–253 μm conservative. No source quantifies incoming RRBO/NMP coalesced drop size here. The d_{32} statistic cannot bound a fine tail, phase inversion does not preserve the old distribution, and “no coalescence credit” is not “no interfacial coalescence duty.”

8 · Reproducibility and preservation

Run `node deliverables/disengager-preliminary-proposal.mjs`. This standalone calculation reads frozen files only, independently solves SN force balance, compares every bottom root against the prior results, and writes this self-contained HTML plus full-precision JSON. No DB/network query, application change, workflow execution, RPM/geometry optimization or Stage-5 overwrite occurs.

Numerical checks: all ten bottom inverse roots match prior values within 10^{-12} m; velocity residuals $<10^{-12}$ m/s; mass closure $<10^{-8}$ kg/h; 900-mm passes and 800-mm fails the assigned 200- μ m screen; elevation ordering and outlet-centre covers checked. These are computational consistency checks, not hydraulic validation.

Frozen input SHA-256: 3e65f398cfd714f599ab7fcb420e07c4a4d3c6637053a71498567099413536d2

Unchanged drag source SHA-256:

9a1def48c229c5af65fe2fe3773f032d556bbdeaccbfe9c49a2166042f34f4ed

Current Stage-1 hash: 06bb9227b30f1f4c67b3ed557db34f61fe749037addf66163194dd46e2e98c5f

Stage-3 ledger 69; Stage-4 row 17; Stage-5 revision 6. Full source-file hashes and provenance are retained in the companion JSON.

Engineering disposition: proceed with this preliminary top/active/bottom layout for review and test planning; do not release fabrication or claim guaranteed separation. No persisted engineering authority has been changed.